Language detection failure

Gareth Tribello[1]

[1]School of Mathematics and Physics

Keywords: Natural language processing, distributions

12th August, 2026

Abstract

An algorithm for distinguishing between German, English, French, Spanish and Italian is designed. This algorithm works uses the fact that each of these languages has a different distribution of letters. We show that if this algorithm is trained using only a single phrase in each of the target languages the differences in the distribution of letters each each language are not statistically significant. In other words, when the computer is only given a phrase of training data you cannot build an algorithm for differentiating between languages.

1 Introduction

Differentiating between text written in Greek and English is straightforward as these two languages use completely different alphabets. However, it is more difficult to differentiate between texts in English and German as the alphabets that are used for both languages are (largely) identical. However, there are noticeable differences ways these different languages use the letters of the alphabet to construct words and phrases. Consequently, in this report we describe how we have tried to build an algorithm for differentiating between different languages by using these differences in the distribution of letters in different algorithms.

We chose to train our algorithm using the following famous quote from Nietzche:

Was mich nicht umbringt, macht mich stärker

This can be translated into various other European languages as follows:

English	That which does not kill me, makes me stronger
French	Ce qui ne me tue pas me rend plus fort
Spanish	Lo que no me mata, me hace más fuerte
Italian	Ciò che non mi uccide mi rende più forte

The report demonstrates that a single phrase does not enough provide enough information for training an algorithm for differentiating between languages. The differences in the distributions of letters in the various phrases are not statistically significant. Furthermore, hypothesis tests performed for the test phrase do not provide sufficient evidence to reject the hypothesis that the phrase is from any one of the five studied languages.

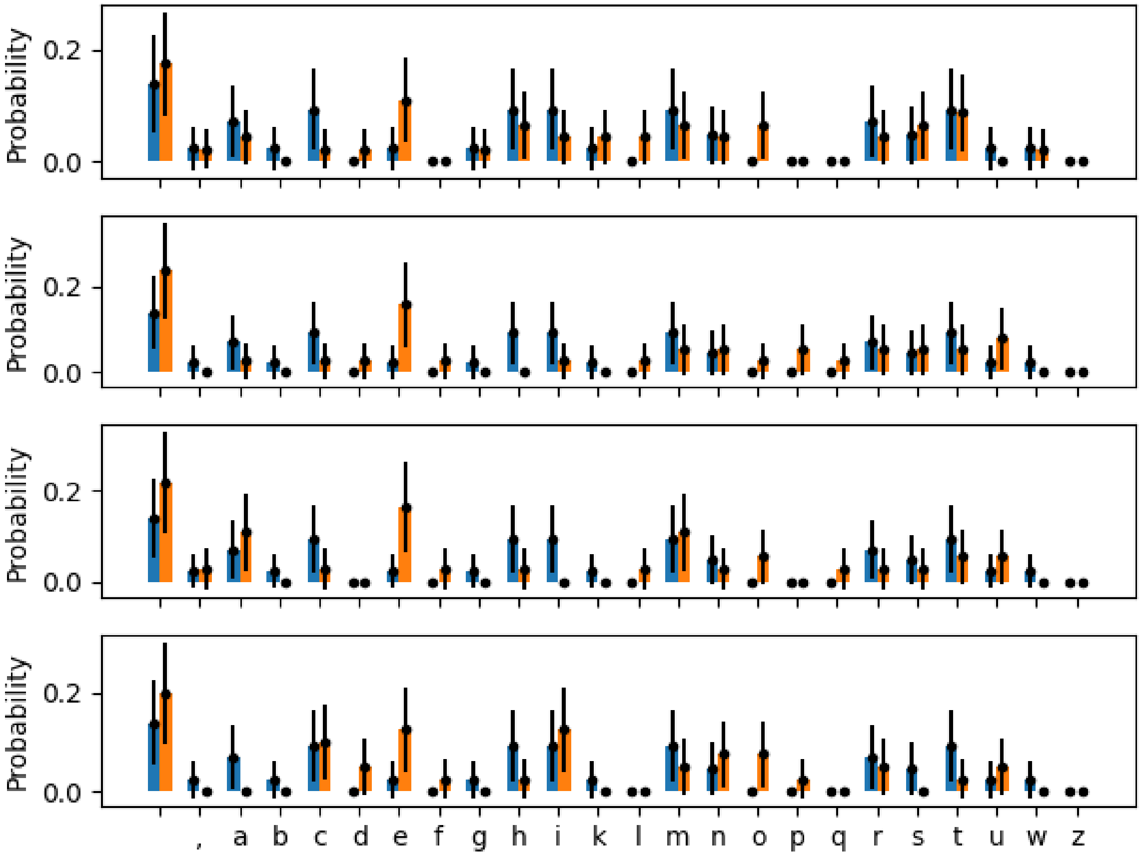

Fig. 1. Comparisons between the distribution of letters in the German version of the training phrase (blue bars) and translations of this phrase (orange bars) in, from top to bottom, English French, Spanish and Italian. Error bars indicate the 90% confidence limit on the estimate of the distribution.

2 Results and discussion

We begun our investigation by constructing a set containing all the symbols that were used to write the training phrase in the five languages and the test phrase in German. The test phrase we used was the German proverb:

Alles hat ein Ende, nur die Wurst hat zwei

which translates into English as: "Everything has an end, only the sausage has two." To ensure that we only used English letters and punctuation marks we removed the accents and umlauts from the non-English phrases. Figure 1 shows comparisons of the distribution of letters in the phrase "was mich nicht umbringt, macht mich stärker" (blue bars) with the distribution of letters in each of the languages we translated this into (orange bars). You can see that, even though some languages use letters that are not employed in the German version of the phrase, the differences between the distributions of letters for the various languages are not statistically significant.

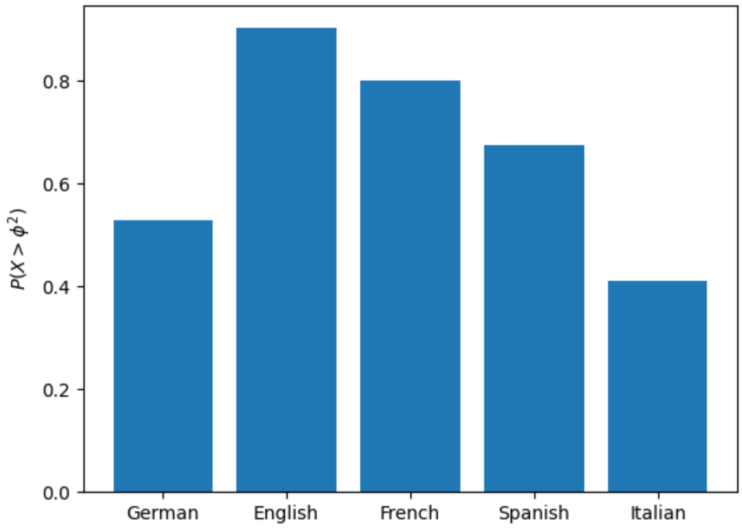

Fig. 2. Results from statistical tests on the homogeneity in the distribution of letters between the German test phrase "alles hat ein Ende, nur die Wurst hat zwei" and translations of the phrase "was mich nicht umbringt, macht mich stärker" in the various languages shown. The figure illustrates that the distribution of letters in the test phrase is consistent with the distribution of letters in each translation of the training phrase.

We tested whether the populations of letters that we obtained from each of the translations of the testing phrase were homogeneous with respect to distribution of letters by computing the test statistic:

$$\phi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \qquad \text{where} \qquad E_{ij} = \frac{\left(\sum_{k=1}^{r} O_{kj}\right)\left(\sum_{l=1}^{c} O_{il}\right)}{\sum_{k=1}^{r}\sum_{l=1}^{c} O_{kl}}$$

In this expression, O_{ij} is the number of observations of letter i in language j and r and c are thus the number of letters and languages respectively. If the probability distributions for the letters in the five languages are all the same then the test statistic above should be a sample from a χ^2 distribution with $(r-1)(c-1)$ degrees of freedom. The probability that a random sample from this χ^2 distribution is greater than the value for the test statistic that we obtained from our data is 0.870. There is thus insufficient evidence to conclude that the distributions of letters for the various languages are different.

That we cannot differentiate between the languages by just looking at the letter distributions in a single phrase is further evidenced by figure 2. To construct this figure we performed five hypothesis tests similar to the one described in the previous paragraph. These hypothesis tests compared the distribution of letters in the German phrase "alles hat ein Ende, nur die Wurst hat zwei" with the distributions of letters in "was mich nicht umbringt, macht mich stärker" translated into the various different languages. The figure shows that there is insufficient evidence to conclude that the distribution of letters in any translation of the second phrase is different from the distribution of letters in the first phrase. In other words, the algorithm is unable to conclude that "was mich nicht umbringt, macht mich stärker" is a German expression. In fact, figure 2 shows that the distribution of letters in the phrase "alles hat ein Ende, nur die Wurst hat zwei" is more similar to the distributions obtained from the English, French and Spanish translations of "was mich nicht umbringt, macht mich stärker" than it is to the distribution from the original German. Only the Italian translation has a distribution of letters that is less similar than the distribution obtained from "alles hat ein Ende, nur die Wurst hat zwei" than the original German.

Thus far we have assumed that the differences in the languages can be captured by examining the proportion of times each of the various letters is used. Obviously, however, these letters are ordered to construct words and phrases in different ways in the various different languages that have been studied here. We thus constructed figure 3 to illustrate how frequency each pair of letters appeared side by side in the original phrase. To construct this figure we constructed a histogram using the $n-1$ pairs of letters in each translation of the phrase "was mich nicht umbringt, macht mich stärker." The resulting histogram has 24×24 bins so, rather than displaying this histogram with frequencies on the y axis we stored the histogram in a matrix. The row index of this matrix indicates the first letter in the pair, while the column index indicates the second letter. Each pixel in the representation of these matrices that are shown in figure 3 is then colored in accordance with the probability of observing the letter pair.

The transition matrices for the languages that are displayed in figure 3 are all sparse. Furthermore, the sparsity patterns for the five different languages are clearly different. Nevertheless, because only a small amount of data was used to construct these histograms a hypothesis test similar to the one described earlier that tests whether the various different homogeneous with respect to distribution of pairs of letters returns a p-value of 1.0, which suggests that there is no reason to believe that all the distributions in 3 aren't all samples of the same underlying distribution.

A similar result is obtained when one compares the distribution of pairs of letters in the phrase "alles hat ein Ende, nur die Wurst hat zwei" with the distribution of pairs of letters from figure 3. The p-value is one for all five of these hypothesis tests, which suggests that there is no reason to believe that this (German) phrase is not in English, French, Spanish or Italian.

The clear visual differences in the sparsity patterns for the language matrices shown in figure 3 suggest an alternative approach for detecting the language of the phrase "alles hat ein Ende, nur die Wurst hat zwei." We can calculate the proportion of letter pairs in "alles hat ein Ende, nur die Wurst hat zwei" that do not appear in the various translations of "was mich nicht umbringt, macht mich stärker." The left-panel of figure 4 shows the result from this analysis. This figure shows that the majority of letter pairs that appear in "alles hat ein Ende, nur die Wurst hat zwei" do not appear in the various translations of "was mich nicht umbringt, macht mich stärker." Furthermore, the matrix that is constructed from the English

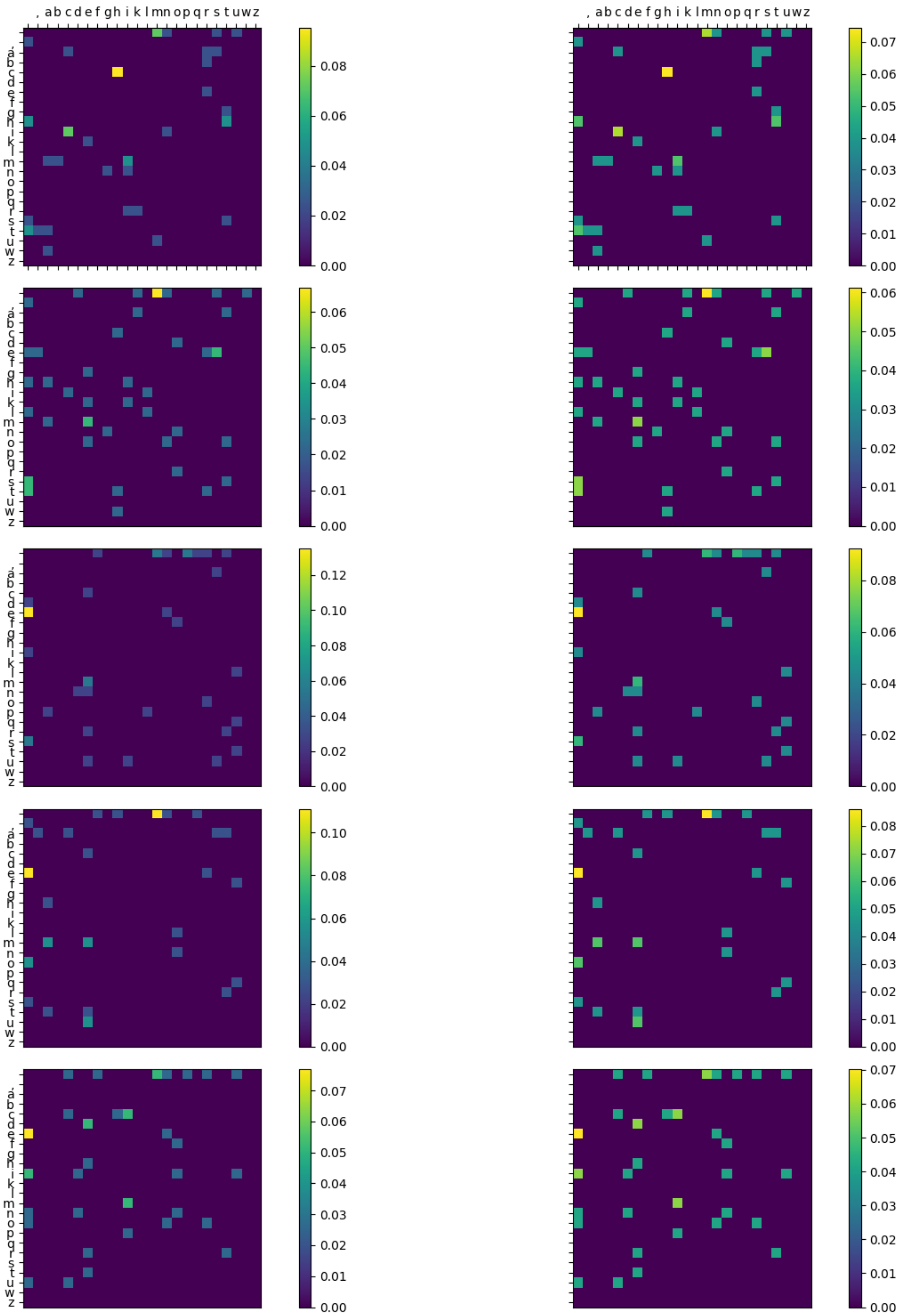

Fig. 3. Transition probabilities between the letters on the left axis and the letters on the top axis in, from top to bottom, German, English, French, Spanish and Italian. These matrices were obtained by looking at the frequency of each pair of letters in the various translations of the phrase "was mich nicht umbringt, macht mich stärker." The figures in the left column show the frequency of each letter pair, while those in the right column are the errors on these estimates of the transition probabilities.

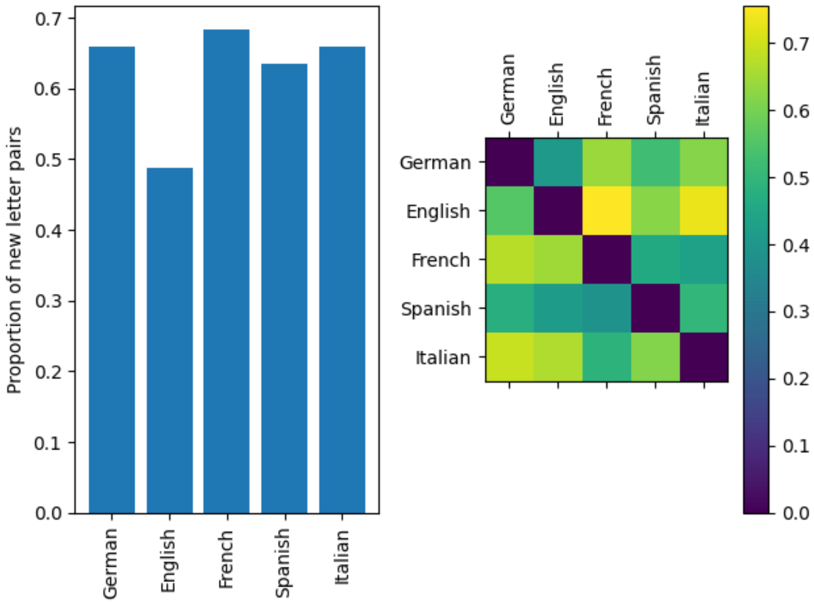

Fig. 4. Left panel: the proportion of letter pairs in the (German) phrase "Alles hat ein Ende, nur die Wurst hat zwei" that do not appear in the various translations of "Was mich nicht umbringt, macht mich stärker." **Right panel** matrix indicating the proportion of letter pairs in the translation of "was mich nicht umbringt, macht mich stärker" in the language shown in the top axis that do not appear in the version of that phrase in the langauge shown on the left axis.

translation of "was mich nicht umbringt, macht mich stärker" has the sparsity pattern that is most similar to the sparsity pattern that is observed in the German phrase about sausages.

For comparison the right panel of figure 4 shows the result obtained when an analysis like that described in the previous paragraph is performed for translations of "was mich nicht umbringt, macht mich stärker" in each pair of languages. Once again we see that the the majority of the letter pairs that appear in one translation of the phrase do not appear in any other translation. Our visual impression from figure 3 was thus correct - the sparsity patterns of the matrices that are shown in that figure all different. However, the proportions of new letter pairs that we observe when we move between languages are similar to the proportion of new letter pairs that are observed when we change from one German phrase to a new and different German phrase. We can thus conclude that a single phrase in German does not provide enough information to characterize the distribution of letters in the German language.

3 Conclusion

This report has demonstrated that you cannot determine the differences between how different languages use the letters of the alphabet to construct words and phrases from a single phrase in each language. There is not enough information in a single phrase to find statistically significant differences between the distributions of letters for different languages.

In theory we could have extended this work by investigating the distribution of sets of three letters but that would reduce the amount of data we have to analyze even further. The next step would thus be to perform the analyses described above with a larger corpus of text in each of the target languages.